

Cezar-Constantin Andrici

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I am a Ph.D. student at MPI-SP, Germany, until December 2026, working with Cătălin Hrițcu on securing verified F^* programs against linked unverified code. Currently, I am also a research intern with Son Ho at Microsoft Research, contributing to verification tools and proof automation for Rust programs. Over the last 8 years, working in F^* , Lean, Dafny and Rocq, I have built verification tools, verified programs, and verified compilers, with a focus on proving security properties.

Education

- 10/2021–12/2026 **Ph.D. student** in Computer Science at **MPI-SP**, Germany
Thesis: *Securing Verified Monadic F^* Programs against Linked Unverified Code*. Advised by C. Hrițcu.
- 10/2019–2021 **M.Sc.** in Computer Science from **UAIC**, Romania
Thesis: *Enforcing trace properties on the interoperability between F^* and ML using hybrid verification*. Advised by Ștefan Ciobâcă and Cătălin Hrițcu.
- 10/2019–2021 **M.A.** in Regional Development from **UAIC**, Romania
Thesis: *Governance, Resilience and Absorption of Structural and Cohesion Funds in Central and Eastern Europe: A Case Study*. Advised by Ramona Țigănașu.
- 10/2015–2019 **B.Sc.** in Computer Science from **UAIC**, Romania
Thesis: *Proving SAT Solving Algorithms and Data Structures in Dafny*. Advised by Ștefan Ciobâcă.

Publications

- Conferences **Misquoted no more: Securely extracting F^* programs with IO.** C.-C. Andrici, D. Ahman, C. Hrițcu, A. Pribisova, E. Rivas, and T. Winterhalter. *ICFP*, 2026
- SecRef*: Securely sharing mutable references between verified and unverified code in F^* .** C.-C. Andrici, D. Ahman, C. Hrițcu, R. Ileanu, G. Martínez, E. Rivas, and T. Winterhalter. *ICFP*, 2025
- Securing verified IO programs against unverified code in F^* .** C.-C. Andrici, Ș. Ciobâcă, C. Hrițcu, G. Martínez, E. Rivas, É. Tanter, and T. Winterhalter. *POPL*, 2024
- Journals **A verified implementation of the DPLL algorithm in Dafny.** C.-C. Andrici and Ș. Ciobâcă. *Mathematics*, 2022
- Informal **Verifying non-terminating programs with IO in F^* .** C.-C. Andrici, T. Winterhalter, and C. Hrițcu. *HOPE*, 2022
- Partial Dijkstra Monads for all,** T. Winterhalter, C.-C. Andrici, C. Hrițcu, K. Maillard, G. Martínez, and E. Rivas. *TYPES*, 2022
- Gradual enforcement of IO trace properties (poster)**
🏆 1st prize at Student Research Competition of ICFP 2020 (graduates section)

Research Projects

- 2026–ongoing Contributor to **Aeneas**, a verification toolchain for Rust programs designed to keep proofs as simple as possible. Extending its Lean backend with separation logic support and proof automation, to verify Rust programs that manipulate raw pointers in unsafe code.
- 2023–ongoing Lead author of **SCIO*** and **SecRef***, two verified secure compilation frameworks for verified IO/stateful F^* programs. Proved preservation of *integrity*, *data confidentiality*, and *code confidentiality* when compiled code is linked against arbitrary adversarial unverified code. The frameworks are themselves implemented and verified in F^* .
- 2022 **TrueSAT:** Implemented a SAT solver in Dafny and verified it to be sound, complete, and terminating. Formalized and proved correctness of the DPLL algorithm.

Experience

08/2026-10/2026	Research intern at Microsoft Research in Security and Privacy Group, advised by Son Ho Contributing to Aeneas , a verification tool chain for Rust programs, by extending the Lean backend with Separation Logic support and automating verification.
02/2020-07/2021	Research intern at UAIC , advised by Ștefan Ciobăcă
07-10/2020	Research intern at MPI-SP in the FOVSEC group, advised by Cătălin Hrițcu
07-11/2019	Research intern at Inria Paris in the Prosecco team, advised by C. Hrițcu and E. Rivas
12/2017-06/2019	System Administrator at UAIC , Romania Responsible for maintaining the servers of the Computer Science Department, providing support to students and staff, and managing the department's IT infrastructure. Migrated the department's 15 year old main website, annals website, and research group website to WordPress.
11/2015-11/2017	Team Lead at CTF365 , Cluj & Remote, Romania Led the technical team for a cybersecurity startup whose VPN-based training platform let users attack target servers across a 150+ server network and served several Fortune 500 companies. Secured pre-seed investment from <i>hub:raum</i> , Deutsche Telekom's tech incubator.
07-10/2015	Software Development Intern at Amazon Development Center , Iași, Romania Built a single-page registration application for an annual external event on top of AWS.

Skills

Verification	F*, Lean, Dafny, Rocq, Iris; secure compilation, program verification, SMT/SAT solving.
P. Languages	Wrote professional software in OCaml, C++, JavaScript, Ruby, SQL, PHP. Other languages include Python, Haskell, Solidity and C#.
Languages	Romanian (native), English (proficient), German (beginner)

Teaching Assistant

2024 Fall	Proofs are Programs, Graduate level course, RUB, Germany
2023 Summer	Functional Programming, Undergraduate level course, RUB, Germany
2020 Fall	Logics in Computer Science, Undergraduate level course, UAIC, Romania

Academic Service

Sub-reviewer	CPP'26, ICFP'25, POPL'24, SP'2021
AEC Member	POPL'26, POPL'23
Student volunteer	POPL'24, POPL'23, ICFP'22, PLDI'20, POPL'20, ETAPS'19, FROM'18
Other	Supporting the CSF 2023-2026 Test of Time Award chairs by automatically gathering and visualizing citation data for over 500 papers each year. Organized electronic elections inside the MPI-SP institute to elect the student representative and Ombudspersons.